

Implementation of Artificial Intelligence in Fashion: Are Consumers Ready?

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Abstract

Given the growing interest in combinations of fashion and digital innovations, it is critical for both researchers and retailers to understand how consumers respond to new technologies, especially artificial intelligence (AI). The purpose of the study was to examine consumers' attitudes and purchase intention toward an AI device. By adapting the technology acceptance model, a conceptual model was constructed and tested related to consumers' attitudes and purchase intention toward an AI device—Echo Look. A total of 313 subjects (61% female) between 18 and 65 years old in the top 10 metropolitan areas in the United States participated in the study. The results indicated that perceived usefulness, perceived ease of use, and performance risk were significant in consumers' attitude toward AI. Positive attitudes toward technology positively influenced the purchase intention. Based on these results, theoretical and practical implications are discussed.

Keywords

fashion artificial intelligence, technology attitudes, purchase intention

Technological advancement has brought dramatic changes to consumers' consumption behaviors. Consumers' self-awareness and fashion sense are being transformed by technology. Because combinations of fashion and digital innovations are emerging, it is critical for researchers, as well as retailers, to understand consumer responses to new technologies (Bues, Steiner, Stafflage, & Krafft, 2017; H.-Y. Kim, Lee, Mun, & Johnson, 2017). For example, in the fashion industry, technologies such as smart in-store technology (H.-Y. Kim et al., 2017), mass customization technology (H.-H. Lee & Chang, 2011), augmented reality (M. Kim & Cheeyong, 2015), virtual try-on (Shin & Baytar, 2014), and virtual personal assistants (VPAs; S. I. Shim, Kwon, Chattaraman, & Gilbert, 2012) are being used by retailers.

As artificial intelligence (AI) has emerged as an important frontier of technological innovation (Hager, Bryant, Horvitz, Mataric, & Honavar, 2017), it has also begun to be used as a new

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application in the fashion industry (Amazon Fashion, 2017; Wong & Liu, 2018). Because AI can deliver significant improvements in speed, cost, and flexibility across the fashion supply chain, it is critical to use AI to reinvent design, merchandising, and marketing (McKinsey & Company, 2018). By analyzing and storing insights from thousands of images and videos using computer vision, AI has assisted designers in integrating trending colors, key patterns, and styles, which can reduce overall lead times and expand designers' creative discovery (Arthur, 2018). AI is also being used to assist stylists with recommending items based on each customer's preferences for price, size, general style, order history, and activities on social media such as fashion pictures saved on Pinterest (Cao, 2018). In addition, AI has been used by fashion retailers (such as Nordstrom, Nike, Macy's, Farfetch) in chatbots, virtual assistants, and product navigators to enhance and integrate consumers' online and off-line experiences (McKinsey & Company, 2018; Sennaar, 2017). One recently launched AI product is Amazon's Echo Look, a virtual style consultant and the latest addition to Amazon's Echo-branded products (Tillman, 2018). Various functions of this product include taking photos via voice commands, a style-check function, and a social sharing function. In the current study, we focus on all the functions of this product, particularly the style-check function, which demonstrates the most advanced technology and is highly promoted in Amazon's commercial video. This new technology combines machine learning and advice from fashion specialists. Currently, Echo Look is not available at retail websites but could possibly be adopted in the future.

According to a recent report from McKinsey & Company (2018), fashion companies are expecting to deploy AI to redefine interactions and engagement with customers as the next frontier. As retailers study consumers' shopping behavior and promote the future of on-demand manufacturing, a better understanding of consumers' acceptance of the application of AI to fashion is needed. However, because the development and application of fashion AI are in the beginning stages, research is still in its infancy as well. Relatively, few researchers have examined and developed studies on consumers' acceptance of fashion AI. With the current study, we aimed to develop and empirically test the technology acceptance model (TAM) in the context of a fashion AI product by incorporating perceived usefulness, perceived ease of use, and perceived risk. Therefore, the purpose of the study was to examine effects of perceived usefulness, perceived ease of use, perceived risk, and attitudes toward technology on consumers' attitudes toward and intention to purchase a fashion AI product and to test the moderating influence of fashion involvement on attitudes toward technology and toward purchase intention of a fashion AI product. More specifically, we intend to shed light on understanding (a) consumers' acceptance and purchase intention toward a fashion AI product, so as to predict how consumers' fashion sense will be affected by new technologies, and (b) the moderating effects of fashion involvement on consumers' acceptance and purchase intention toward a fashion AI product. A conceptual model based on TAM was developed, and a set of research hypotheses consistent with research objectives was proposed based on the conceptual model. By analyzing what belief is the best predictor of adoption intention of a technology innovation, we will provide theoretical and managerial implications for future technology development. Research findings will be beneficial for industry professionals to better understand consumer preferences and acceptance with regard to the application of AI in the fashion industry.

Literature Review and Hypotheses Development

The technology acceptance model (TAM), which was adapted from the theory of reasoned action (TRA), is specifically tailored for examining acceptance of information systems. The goal of TAM is to provide an explanation of the determinants of computer acceptance across a broad range of end-user computing technologies and user populations, while at the same time being both parsimonious and theoretically justified (Davis, Bagozzi, & Warshaw, 1989). "A key purpose of TAM, therefore,

is to provide a basis for tracing the impact of external factors on internal beliefs, attitudes, and intentions” (p. 985).

TAM posits that two particular beliefs, perceived usefulness and perceived ease of use, are of primary relevance for technology acceptance (Davis et al., 1989). Even in the original TAM, the researchers viewed perceived usefulness and ease of use as related constructs. It was found in recent studies related to technology in general (Leong, Ibrahim, Dalvi-Esfahani, Shahbazi, & Nilashi, 2018; Yang & Wang, 2019) that a relationship between the two constructs was not supported. The majority of respondents were tech-savvy individuals between 25 and 44 years old, so they may have considered the product easy to use. Therefore, perceived ease of use may not theoretically lead to users’ corresponding perceived usefulness. To keep consistent with recent research and considering the characteristics of participants, we did not include this relationship in our proposed model. Similar to TRA, TAM postulates that computer usage is determined by behavioral intention (BI) to use, which is described as an individual’s cognitive plan to use an information system (Davis et al., 1989). In marketing research, BI was conceptualized as consumers’ willingness to purchase, and it is established by an individual’s attitude toward the information system (S. Shim, Morris, & Morgan, 1989).

TAM has been widely used to assess consumers’ acceptance of technology-related applications in the fashion industry. The model has been used to test consumers’ attitudes toward and acceptance of online mass-customization technology (H.-H. Lee & Chang, 2011), wearable fitness technology (Lunney, Cunningham, & Eastin, 2016), and smart in-store technology applications in retailing (H.-Y. Kim et al., 2017), confirming positive relationships between perceived usefulness, perceived ease of use, and acceptance. Researchers have extended the original TAM into different contexts, as well as identifying and including additional variables. Thus, apart from the two primary beliefs, perceived usefulness and perceived ease of use, performance risk (H.-H. Lee & Moon, 2015) and technology attitudes (Rosen, Whaling, Carrier, Cheever, & Rokkum, 2013) were added to extend the original TAM. In this study, fashion involvement was added to test its moderating effects across groups of higher (vs. lower) fashion involvement. The conceptual framework in Figure 1 shows variables included and proposed hypotheses.

Perceived Usefulness

Perceived usefulness is “the degree to which a person believes that using a particular system would enhance his or her job performance” (Davis, 1989, p. 320). Perceived usefulness in TAM has been found to have a significant impact on attitudes toward using the technology (J. Kim & Forsythe, 2007; Williams, Slade, & Dwivedi, 2014). Echo Look is a product in which the embedded hands-free camera enables users to take photos or videos of themselves (“selfies”). The style-check function of the product provides consumers with professional suggestions that will improve their performance in choosing the trendiest outfit. Moreover, the built-in lighting and depth-sensing camera provide users with clear, full-length photos, and videos for instantly sharing on social media. With these functions and interactions, consumers’ efficiency in choosing the trendiest outfit will be enhanced, and these functions can make it easier for consumers to choose what to wear. Therefore, consumers will likely have a positive attitude toward the new fashion AI product. Therefore, we propose:

Hypothesis 1: Perceived usefulness will positively influence consumers’ attitude toward AI.

Perceived Ease of Use

Perceived ease of use is an important determinant of use of technology or systems (Davis, 1989, 1993; Davis, Bagozzi, & Warshaw, 1992; Mathieson, 1991). The importance of perceived ease of

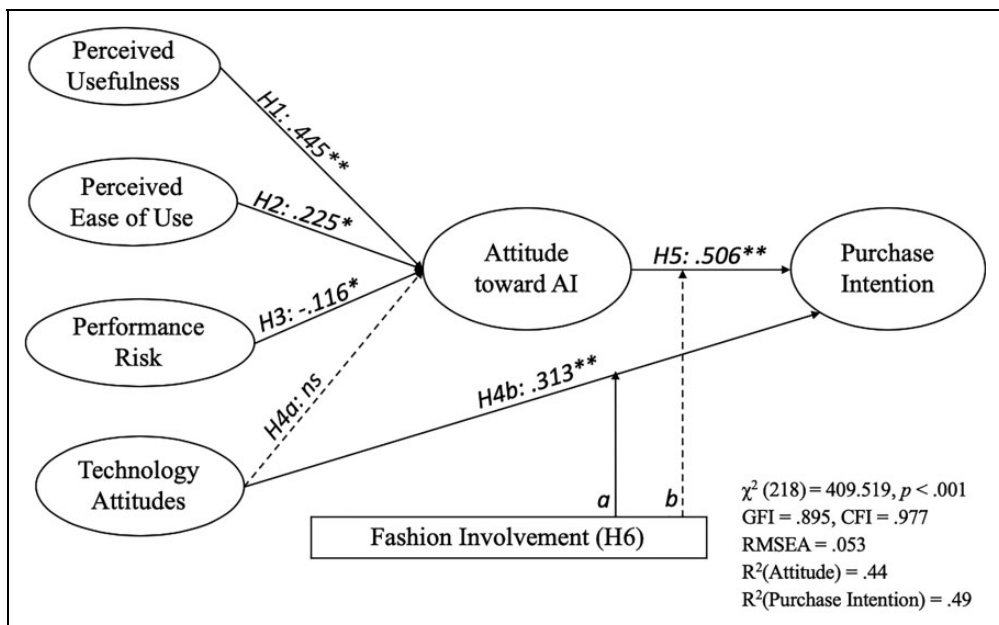


Figure 1. Conceptual framework and testing results.

use has been highlighted in TAM because of the impact a poor user interface has on rejection of IT technology (Venkatesh & Davis, 1996). Perceived ease of use in TAM is “the degree to which a person believes that using a particular system would be free of effort” (Davis, 1989, p. 320). Prior researchers validated a positive relationship between perceived ease of use and consumers’ attitude toward technology. H.-Y. Kim, Lee, Mun, and Johnson (2017) found that perceived ease of use is a significant factor in determining adoption of smart retail technologies, and Lunney, Cunningham, and Eastin (2016) indicated that perceived ease of use positively influenced consumers’ attitude toward wearable fitness technology. Ease of use for the AI product means that it is easy to operate and interact with the program. By employing the Alexa voice control function, consumers interact with the product (i.e., Echo Look) by speaking. Brill (2018) indicated that improved consumer satisfaction will result if digital assistants such as Alexa can meet consumers’ expectations. Therefore, if it is easy for consumers to operate and interact with this product, they will likely develop a positive attitude. Therefore, we propose:

Hypothesis 2: Perceived ease of use will positively influence consumers’ attitude toward AI.

Performance Risk

Perceived performance refers to an individual’s subjective assessment about the performance of a product’s attributes, levels of attributes, or outcomes (Spreng & Olshavsky, 1992). It has been defined as “the possibility that the product will not function as expected and/or will not provide the desired benefits” (Grewal, Gotlieb, & Marmorstein, 1994, p. 145). Performance risk refers to the possibility that a product proves disappointing when it does not match the purchaser’s expectation (H.-H. Lee & Moon, 2015). Prior researchers found that performance risk negatively influenced attitude (Hwang, Chung, & Sanders, 2016) and purchase intention (J. Park & Stoel, 2005). Thus, if consumers are concerned that the product will not perform the way they expected (e.g., failed to

fulfill the voice command functions and style-check functions), their expectations will not be met, and a negative attitude will probably result. Therefore, we propose:

Hypothesis 3: Performance risk will negatively influence consumers' attitude toward AI.

Technology Attitudes

Previous researchers indicated that technology tends to trigger both positive and negative feelings, which result in anxiety (Lin & Hsieh, 2006). J. Kim and Forsythe (2009) indicated that technology anxiety (i.e., anxiety related to dependence or addiction to technology) negatively impacted consumers' use of sensory-enabling technology for online apparel shopping. Prior researchers (Curran, Meuter, & Surprenant, 2003) indicated that attitudes toward technology in general are positively related to favorable attitudes or willingness to adopt a technological product. Lin and Hsieh (2006) also indicated that the more favorably an individual feels about technology in general, the more he or she is likely to have favorable attitudes or greater willingness to purchase new technology products. Thus, more positive attitudes toward technology in general can result in more favorable attitudes toward a new technological product and subsequently greater purchase intention. Therefore, Hypothesis 4 was proposed:

Hypothesis 4: Attitudes toward technology in general will positively influence consumers' (a) attitudes and (b) purchase intention toward an AI product.

Attitude Toward AI

Attitudes are lasting, overall evaluations of people, objects, or issues (Baron & Byrne, 1987). Ajzen and Fishbein (1980, p. 6) defined attitude as an "individual's positive or negative evaluation of performing the behavior." In this study, "attitudes toward AI" refers to the specific product, Echo Look. Particularly, it refers to attitudes toward the product performance of related functions. Previous researchers have confirmed the influence of consumers' attitudes on purchase intention (Belleau, Summers, Xu, & Pinel, 2007; Wu, Wu, & Chang, 2016). Thus, we propose:

Hypothesis 5: Consumers' attitude toward AI will positively influence their purchase intention toward the AI product.

Fashion Involvement

Involvement is a measure that can be used to examine consumer behavior and to segment consumer markets (E. J. Park, Kim, & Forney, 2006). It determines one's tendency to pay close attention to products or to engage actively in particular product acquisition activities (H. Kim, 2008). Fashion involvement was defined as "the extent to which consumers view fashion related activities as a central part of their life" (O'Cass, 2004, p. 870). Fashion involvement has been used primarily to predict behavioral variables related to apparel products such as product involvement, buying behavior, and consumer characteristics (Browne & Kaldenberg, 1997; Fairhurst, Good, & Gentry, 1989). Previous researchers found fashion involvement had moderating effects when studying consumers' shopping behavior (J.-H. Lee & Im, 2008). J.-H. Lee and Im (2008) indicated that different levels of fashion involvement presented different effects on perceived justice related to postcomplaint behavior. Consumers of higher level fashion involvement value fashion clothing more, are less price-sensitive, and are more engaged in fashion-related activities such as seeking out fashion news and

trying new fashion-related technologies (Naderi, 2013; Yanshu & Guo, 2017). Celik and Kocaman (2017) concluded that fashion involvement has significant and direct influences on technology readiness. It is assumed that consumers with higher fashion involvement will be more likely to enjoy the interaction with the AI product. Therefore, we propose:

Hypothesis 6: The effects of (a) technology attitudes and (b) attitudes toward AI on purchase intention are stronger for consumers of higher (vs. lower) fashion involvement.

Research Method

Participants

This research was conducted in the top 10 metropolitan areas in the United States. An online self-administered questionnaire was created using Qualtrics, and a subject pool was recruited via Qualtrics Panel services. A total of 313 valid responses were obtained over a 1-week period.

Procedure

Participants viewed a 30-s commercial video advertising Amazon's Echo Look (Amazon Fashion, 2017), and then provided answers to items in the questionnaire, which included measures from established research on perceived usefulness, perceived ease of use, performance risk, technology attitudes, attitude toward AI, purchase intention, and fashion involvement. Items of fashion involvement were accompanied by 5-point scales; all other items were accompanied by 7-point scales. The questionnaire took about 15 min to complete.

Echo Look was used as the stimulus. It was designed by Amazon as an approach to AI from the end user's (consumer's) perspective. The product was launched by invitation in April 2017 and opened to general customers in June 2018 (Eadicicco, 2017). Echo Look is a product that combines fashion AI, machine learning, the Alexa voice control for Amazon Echo (a product that is installed in the home and controls connected functions via voice commands), and a camera (Applin, 2017). Echo Look also serves as a virtual-style consultant, enabling a person to check what they are wearing to see whether it is stylish and aesthetically pleasing (Applin, 2017). Consumers can use the Echo Look (with built-in lighting and depth-sensing camera) to take photos and videos by using the voice command and share instantly with friends (Amazon Fashion, 2017). Moreover, the style-check function (which combines machine learning and advice from fashion specialists) provides consumers with fashion suggestions when they are comparing two styles (Amazon Fashion, 2017). The video link is provided in Table 1.¹

Research Instruments

A comprehensive review of literature was conducted to search for quality measures for the research constructs in the proposed model (see Table 1). Measures were adopted or adapted from previous research based on validity and reliability.

Analysis

For data analysis, descriptive statistics, reliability, structural equation modeling (SEM), and multi-group comparison were used with SPSS 25 and Amos 25 statistical software programs.

Table 1. Research Measurement and Its Source.

Construct	Variable Type	Source	Format After Adapted
Perceived usefulness	Exogenous variable	Cho and Fiorito (2009)	6-Item scale; 7-point Likert-type (<i>very unlikely/very likely</i>)
Perceived ease of use	Exogenous variable	Cho and Fiorito (2009)	6-Item scale; 7-point Likert-type (<i>very unlikely/very likely</i>)
Performance risks	Exogenous variable	H.-H. Lee and Moon (2015)	3-Item scale; 7-point Likert-type (<i>strongly disagree/strongly agree</i>)
Technology attitudes	Exogenous variable	Rosen, Whaling, Carrier, Cheever, and Rokkum (2013)	4-Item scale; 7-point Likert-type (<i>strongly disagree/strongly agree</i>)
Attitude toward AI	Endogenous variable	Batra and Ahtola (1991)	5-Item scale
Purchase intention	Endogenous variable	Dodds, Monroe, and Grewal (1991)	3-Item scale with the end points (<i>very low to very high</i>)
Fashion involvement	Moderating variable	O'Cass (2004)	11-Item scale; 5-point Likert-type (<i>strongly disagree/strongly agree</i>)

Note. The link to the commercial video of Echo Look (https://www.youtube.com/watch?v=9X_fP4pPWPw). AI = artificial intelligence.

Results

Sample

The demographic characteristics of the sample are shown in Table 2. The majority of respondents were female (61%), and 50.8% ranged between 25 and 44 years old (range = 18–65). Participants were categorized into two fashion involvement groups (low, $n = 163$; high, $n = 150$) based on median scores.

Exploratory Factor Analysis (EFA)

EFA using a principal component analysis method was performed on 27-scale items of the exogenous variables and endogenous variables. Because of high cross loading ($>.40$; Hair, Black, Babin, & Anderson, 2010), 1 item of the technology attitudes was dropped. The final factor analysis solution had a total of 26 items that measured six factors and accounted for approximately 85.06% of the total variance. All commonalities ranged between .666 and .966, while Cronbach's α ranged from .785 to .982, demonstrating the good reliability of the scales. EFA loadings ranging from .748 to .932 are reported in Table 3.

Confirmatory Factor Analysis (CFA)

CFA was conducted on perceived usefulness, perceived ease of use, performance risk, technology attitudes, attitude toward AI, and purchase intention of Echo Look. Three items with high modification indices were dropped from further analysis. Dropped items included one from perceived usefulness and two from perceived ease of use. CFA on all remaining 23 items showed an excellent fit ($\chi^2 = 339.827$; $df = 215$; $\chi^2/df = 1.581$; $p < .001$; root mean square residual [RMSEA] = .043; comparative fit index [CFI] = .985; goodness-of-fit index [GFI] = .913), providing evidence of convergent validity. The good fit indices lend support for the construct validity of individual constructs in the model, as indicated by the earlier EFA.

As reported in Table 4, each item loaded significantly on its proposed constructs, with composite reliabilities above .80, providing evidence of the reliability of the measures (Hair et al., 2010).

Table 2. Demographic Characteristics of Research Sample.

Characteristic	Percent	Characteristic	Percent
Gender		Education	
Male	39.3	Less than high school	1
Female	60.7	High school graduate	12.7
Age		Some college	22.7
18–24	8.9	2-Year degree	9.3
25–34	28.2	4-Year degree	37.7
35–44	22.7	Professional degree	14.7
45–54	15.6	Doctorate	1.9
55–64	23.6	Total household income	
65 and above	1.0	Less than \$5,000	1.9
Ethnicity		\$5,000–9,000	2.9
Caucasian	72.2	\$10,000–19,999	5.1
African American	11.2	\$20,000–29,999	10.5
Asian/Asian American	9.9	\$30,000–39,999	8.3
Hispanic/Latino	6.1	\$40,000–49,999	9.3
Native American	0.3	\$50,000–59,999	10.5
Other	0.3	\$60,000–69,999	9.3
Employment		\$70,000–79,999	7.0
Employed full time (40 or more hr per week)	47	\$80,000–89,999	6.1
Employed part time (up to 39 hr per week)	9.9	\$90,000–99,999	4.5
Unemployed and currently looking for work	6.1	\$100,000–149,999	14.4
Unemployed and not currently looking for work	1.9	\$150,000–199,999	6.4
Graduate student	1	\$200,000–249,999	2.2
Undergraduate student	2.2	\$250,000 or more	1.6
Retired	8	Marital status	
Homemaker	15.7	Married	54.0
Self-employed	3.7	Single	42.2
Unable to work	4.5	Other	3.8

Note. $n = 313$.

Results showed good internal consistency of multiple indicators for each construct. The average variance extracted (AVE), which ranged from .559 to .948, exceeded the recommended value of .50 (Fornell & Larcker, 1981). All standardized CFA loadings were significant ($p < .001$) and exceeded .70 (ranging from .745 to .982), showing strong convergent validity (items that are indicators of a specific construct share a high proportion of variance in common; Anderson & Gerbing, 1988; Hair et al., 2010). Also, as shown in Table 5, AVE for each construct was greater than the estimates of squared correlations between constructs, confirming discriminant validity (the extent to which a construct is truly distinct from other constructs; Fornell & Larcker, 1981; Hair et al., 2010).

Model Development and Hypotheses Testing

SEM (Hair et al., 2010; Kline, 2010) was used to test the research model. The model fit was very good ($\chi^2 = 409.591$; $df = 218$; $\chi^2/df = 1.879$; $p < .001$; RMSEA = .053; CFI = .977; GFI = .895). A comparison of these values against recommended values suggests that the model estimation result is satisfactory (Kline, 2010; MacCallum, Browne, & Sugawara, 1996). Results of hypothesized relationships are summarized in Table 6. As shown in Table 6, perceived usefulness, perceived ease of use, performance risk, and technology attitudes positively influence consumers' attitude toward AI. Attitude toward AI had a direct positive effect on consumers' purchase intention. Therefore, we

Table 3. Exploratory Factor Analysis (EFA) Results.

Factor	Scale Item	EFA Loadings	Reliability
Perceived usefulness (PU)	Choose my outfit faster (PU 1)	.834	.980
	Improve my performance in choosing the most trendy outfit (PU 2)	.862	
	Increase my efficiency in choosing the most trendy outfit (PU 3)	.864	
	Enhance my effectiveness in choosing the most trendy outfit (PU 4)	.866	
	Make it easier for me to pick out what to wear (PU 5)	.841	
	Be useful for choosing the most trendy outfit (PU 6)	.860	
Perceived ease of use (PEOU)	Learning to operate this device would be easy (PEOU 1)	.810	.953
	I think I would find it would be easy to get this device to do what I want it to do (PEOU 2)	.801	
	My interaction with this device would be clear and understandable (PEOU 3)	.822	
	This device would be flexible to interact with (PEOU 4)	.803	
	It would be easy to become skillful at using this device (PEOU 5)	.862	
	I think this device would be easy to use (PEOU 6)	.857	
Performance risk (PR)	I am concerned that the product advertised in the video is different from the actual product (PR 1)	.891	.894
	I am afraid that the product advertised in the video would not perform the way I expect (PR 2)	.932	
	I am concerned about whether this product would really “perform” as well as it is supposed to (PR 3)	.863	
	Technology will provide solutions to many of our problems (TechAtt 2)	.779	
Technology attitudes (TechAtt)	With technology anything is possible (TechAtt 3)	.756	.785
	I feel that I get more accomplished because of technology (TechAtt 4)	.748	
	Worthless—Valuable (Attitude 1)	.761	
Attitude toward AI	Unfavorable—Favorable (Attitude 2)	.859	.949
	Disagreeable—Agreeable (Attitude 3)	.834	
	Harmful—Beneficial (Attitude 4)	.861	
	Dislike—Like (Attitude 5)	.801	
	The likelihood that I would purchase Echo Look (PI 1)	.808	
Purchase intention (PI)	The probability that I would consider buying Echo Look (PI 2)	.785	.982
	My willingness to buy Echo Look (PI 3)	.789	

Note. AVE = average variance extracted; AI = artificial intelligence.

found support for Hypotheses 1, 2, 3, 4b, and 5. We do not find support for Hypothesis 4a, in which we suggested that technology attitude positively influences consumers' attitude toward AI.

Testing Moderating Effects

The sum of respondents' evaluation on all 11 items of fashion involvement was calculated, and the median score (33.00) was used to form two groups. The items were obtained from O'Cass's (2004) study, which evaluated how much participants are interested in fashion clothing (e.g., “I am very interested in fashion clothing”) and how much they value the importance of fashion clothing in their

Table 4. Confirmatory Factor Analysis (CFA) Results of Measurement Properties.

Scale Item	Composite Reliability	AVE	CFA Loadings
Perceived usefulness (PU)	.977	.896	
PU 1			.893
PU 2			.953
PU 3			.966
PU 4			.957
PU 6			.961
Perceived ease of use (PEOU)	.945	.811	
PEOU 2			.876
PEOU 3			.949
PEOU 4			.899
PEOU 5			.877
Performance risk (PR)	.899	.750	
PR 1			.818
PR 2			.978
PR 3			.790
Technology attitude (TechAtt)	.791	.559	
TechAtt 2			.822
TechAtt 3			.669
TechAtt 4			.745
Attitude toward AI	.952	.800	
Attitude 1			.766
Attitude 2			.946
Attitude 3			.929
Attitude 4			.900
Attitude 5			.919
Purchase intention (PI)	.982	.948	
PI 1			.962
PI 2			.982
PI 3			.977

Note. AI = artificial intelligence.

Table 5. Squared Correlation Matrix With AVE on the Diagonal.

Variables	Attitude Toward AI	Perceived Usefulness	Technology Attitude	Performance Risk	Purchase Intention	Perceived Ease of Use
Attitude toward AI	.894					
Perceived usefulness	.610	.946				
Technology attitude	.439	.536	.748			
Performance risk	-.256	-.161	-.125	.866		
Purchase intention	.636	.716	.492	-.147	.947	
Perceived ease of use	.546	.608	.600	-.282	.502	.901

Note. Values along the diagonal indicate the average variance extracted for each construct. Off-diagonal values indicate squared correlations between constructs. AVE = average variance extracted; AI = artificial intelligence.

life (e.g., “Fashion clothing is an important part of my life”). Multivariate analysis of variance showed that consumers of higher fashion involvement were younger and had more stable occupations than consumers of lower fashion involvement, but there was no difference in ethnicity, gender, education, and income between the groups. Multigroup comparison was conducted to examine

Table 6. Summary of Hypotheses (1–5) and Testing Results.

Path to	Path From	Path Coefficient	<i>p</i>	Hypothesis Testing
Attitude toward AI	Perceived usefulness	.445	**	Hypothesis 1 supported
	Perceived ease of use	.225	*	Hypothesis 2 supported
	Performance risk	−.116	*	Hypothesis 3 supported
	Technology attitudes	.032	<i>ns</i>	Hypothesis 4a not supported
Purchase intention	Technology attitudes	.313	**	Hypothesis 4b supported
	Attitude toward AI	.506	**	Hypothesis 5 supported

Note. AI = artificial intelligence.

p* < .05. *p* < .001.

whether the magnitude of the influences from perceived usefulness, perceived ease of use, performance risk, technology attitudes on attitude toward AI, and the magnitude of the influence from attitude toward AI on purchase intention differed across these two groups (Hair et al., 2010).

A constrained multigroup model (Model 1/base model—no moderating effects) was estimated. Each structural weight was constrained to be equal across the two groups. This model had an acceptable fit ($\chi^2 = 754.404$; $df = 447$; $\chi^2/df = 1.688$; CFI = .959; RMSEA = .047). An unconstrained multigroup model (Model 2—moderating effects) was then estimated, in which the structural weights were estimated uniquely for each group. The unconstrained model exhibited an acceptable fit ($\chi^2 = 657.128$; $df = 436$; $\chi^2/df = 1.507$; CFI = .971; RMSEA = .04). Therefore, the χ^2 difference ($\Delta\chi^2 = 97.276$; $df = 11$; $p < .001$) between the two models was significant at the group level, indicating the influences from perceived usefulness, perceived ease of use, performance risk, technology attitudes on attitude toward AI, and the magnitude of the influence from attitude toward AI on purchase intention do differ for consumers of high fashion involvement and low fashion involvement. To further test the influence from each exogenous variable, each path was constrained separately and the χ^2 difference was compared with the χ^2 threshold. The results indicate that the relationship between technology attitude and purchase intention is significantly different (with 99% confidence) for consumers of high versus low fashion involvement. Specifically, the effect of technology attitude on purchase intention is significant only for consumers of higher fashion involvement. Therefore, Hypothesis 6a was supported, and Hypothesis 6b was not supported. The testing results are summarized in Figure 1.

Discussion

By adapting the research measurement of TAM (Davis, 1989), a conceptual model was constructed and tested related to consumers' attitudes and purchase intention toward an AI product—Echo Look. First, consistent with the results of previous studies (H.-Y. Kim et al., 2017; Li & Huang, 2009), we discovered that perceived usefulness and perceived ease of use were significant in predicting consumers' attitude toward AI. A reasonable explanation is that consumers appreciate the functions that were promoted by the AI product. With these functions and the user-friendly system, consumers' efficiency in selecting trendy outfits was highly enhanced, which led to a more positive attitude toward AI. Moreover, based on the structural model results, we learned that performance risk was negatively significant in predicting consumers' attitude toward AI. If the product failed to fulfill the voice command functions and style-check functions, consumers would not obtain satisfaction and enjoyment from this product, which would lead to a negative attitude toward AI. Interestingly, this also suggested that technology attitudes positively influenced purchase intention but did not influence attitudes toward AI. A logical extension of this thinking is that consumers who are more

positive about the technology would be more likely to try new technologies, even though they may not have established any concrete attitude about a particular technological product. Similar to Belleau, Summers, Xu, and Pinel (2007), H.-H. Lee and Chang (2011), Li and Huang (2009), and others, we confirmed the positive influence from attitude toward AI to purchase intention.

In our results, it was shown that fashion involvement moderated the path from technology attitudes to purchase intention, but not the path from attitudes toward AI to purchase intention. These results are consistent with previous studies in that consumers of higher fashion involvement are more likely to be ready to use new technology (Celik & Kocaman, 2017); however, they are not quite ready to purchase the fashion AI product. Perhaps they are waiting for more customer reviews and a newer version with improved functions or performance of this one of a kind product.

Implications

The results of the study have theoretical and practical implications. The study theoretically contributed to the TAM by extending the model to consumers' evaluation of a new fashion technology. Moreover, technology attitudes were incorporated within the TAM framework, and in our test results, a positive influence on purchase intention was indicated. Addition of these variables provided insights for future research about the application of AI to fashion. In addition, the TAM was applied to a national sample, which can contribute to the existing TAM literature.

From a managerial perspective, the structure and results of this study provided a systematic strategic improvement plan. First, we demonstrated that consumers' evaluation of perceived usefulness and perceived ease of use significantly influence their attitude toward AI, which confirmed findings from previous researchers that functions and interactions with technology enhanced consumers' expectations and satisfaction (H.-Y. Kim et al., 2017; Wu et al., 2016). Retailers could constantly upgrade the platforms of this new fashion technology product to keep it in a user-friendly interface. Currently, the product helps users take full-length pictures but can only take one picture/video at a time. Further development could provide users the option to zoom in and cut the photos and the option to take "burst mode" photos. Adding a front screen/mirror could also improve the interaction, which may generate a higher consumer acceptance level. Secondly, performance risk negatively influenced consumers' attitude toward AI. Launched in 2017 (Amazon Fashion, 2017), Echo Look is still very new to public consumers and can only be purchased by invitation. Moreover, because fashion AI was introduced only at the company level and has not yet reached the customer level, there are no similar fashion AI products on the market with which to compare Echo Look. Therefore, consumers have concerns about the performance of this product. Increasing the presence and demonstration of this product in fashion shows, trade shows, magazines, and social media would decrease consumers' risk concerns. Moreover, future updates could include improved camera quality and variation of angles so as to better meet consumers' expectations.

In addition, instead of being limited to personal use at home, this type of fashion AI product can be developed for an in-store setting as a means to decrease consumers' risk concerns and enhance their knowledge of fashion AI. An in-store setting might provide an opportunity to promote sales, help alleviate consumer indecision/anxiety about which product is best, demonstrate how easy it is to use, explain where the product gets its information, and so forth. A joint special event hosted by a retailer and the company that sells Echo Look could target customers who have indicated an interest and involvement with such fashion events. For example, compared with other customers, these customers have indicated that fashion clothing is a significant part of their lives by having a store credit card, shopping more often, spending more money, and attending more fashion-related events. An invitation-only event could target customers who have the store credit card.

Lastly, fashion involvement would be a good way for retailers to identify potential customers because customers of high fashion involvement are more likely to try new fashion-related

technology and products than customers of low fashion involvement. When promoting this one of a kind product, retailers should reach out to customers of high fashion involvement first. This group of customers can be reached through their current consumer profile (by asking them to fill out a short questionnaire) to select those of higher fashion involvement and through public relations by contacting fashion innovators on social media.

Limitations and Future Research

The current study has some limitations that suggest interesting opportunities for future research. This study recruited participants from the top 10 metropolitan areas in the United States, which may have limited the representativeness of the results. Future researchers may explore attitude and purchase intention for consumers from both urban and rural areas. Moreover, as this study used the Echo Look as stimulus and consumers did not have the experience of using it before, research findings need to be interpreted cautiously due to dynamic and ever-changing technology advancement. This study's focus on a single product (i.e., Amazon's Echo Look) limits its generalization to other fashion AI products. Future researchers may extend this current model to other types of fashion AI products (such as in-store technology and VPAs) and compare the perceptions with consumers' postadoption evaluations.

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Note

1. The following is a verbatim transcript of the video shown to participants:

"Alexa helps with thousands of things, and now she can help you to look your best. 'Alexa take a photo.' Introducing Echo Look, a first of its kind Echo with hands-free camera. Echo Look takes photos using just your voice. Its built-in lighting and the depth-sensing camera blur the background to make sure your outfits pop, giving you clean, full-length photos that are easy to share with friends. Plus, get a live view or take videos to see yourself from every angle. 'Alexa take a video.' We've also created an easy way to get a second opinion. Introducing style-check. It combines the best machine learning with advice from fashion specialists. Just pick two outfits and style-check will give you recommendations based on current trends and what flatters you. Alexa can also help you to create a personal look-book. It shows you what you wore and when. So, you can keep track with your favorites and take your closet with you, wherever you go."

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